SAFe®
PROVIDED BY SCALED AGILE

SAFe® Explained

Succeeding with Lean and Agile at Scale

Work Differently. **Build the Future.**

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PART ONE

Executive Summary

All organizations need to move quickly and improve how they innovate to win. The best way to do this is to adopt Lean-Agile practices that are proven to work. Many Harvard Business Review articles, books, and analysts like Gartner agree that adopting Lean-Agile improves innovation, reduces time to value, and delivers better products at lower cost.

However many organizations struggle to adopt Agile. Some have adopted the Agile labels, but still function in a traditional manner. Achieving the full benefits of Lean and Agile requires changing how the organization operates and requires leadership to adopt new behaviors. Leadership needs to set objectives but avoid weighty documentation and micro-management of what to do, how to do it, and when. This can seem daunting, but there are ways to leverage best practices by implementing the proven model of the Scaled Agile Framework® (SAFe), which has been adopted overwhelmingly by the world's leading organizations to improve their results.

This eBook explains SAFe and why modern organizations leverage it to thrive. With over two million trained professionals and adoption by 20,000+ organizations — including seventy of the Fortune 100 — SAFe has become a proven approach for change, setting a new standard for achieving agility at scale.

Why read this eBook? Are you experiencing some of these problems?

- **We keep failing at Agile.** Many medium and large organizations struggle with Agile because they lack a structured approach for scaling to multiple teams and departments. In the section *Getting Started: A Proven Method for Scaling Change*, we explore why past efforts have failed and how SAFe provides a clear, repeatable path to success.

- **We struggle to quantify the benefits of our Agile implementation.** SAFe isn't just a theory—it's been successfully implemented in some of the world's most complex organizations. In the *Benefits of SAFe* section, we highlight real-world examples of how it helps organizations navigate complexity while driving measurable business outcomes.

- **Our organization thinks frameworks are too rigid.** SAFe offers a flexible framework tailored to your unique context. Instead of spending years developing an unproven operating model with custom training materials and job aids, organizations can adapt SAFe to suit their specific needs. This approach enables a focus on execution and value delivery while building on a proven, scalable foundation for agility. SAFe continuously evolves to incorporate the latest best practices and insights, ensuring organizations stay ahead of industry trends.

SAFe's widespread adoption is driven by its proven ability to enhance efficiency and boost employee morale. It also reduces costs, accelerates time to market, and drives improvements in revenue and quality. These benefits make SAFe the best option for large, complex organizations seeking to scale Lean and Agile practices effectively.

Recognizing its impact, Gartner named SAFe the #1 most considered and adopted framework for scaling Agile.[1] The 17th Digital.ai State of the Agile Survey reports that SAFe is the most popular, with 26% identifying it as the framework they most closely follow for scaling Agile.

SAFe is also built into leading solutions from our software partners including:

servicenow. BROADCOM ATLASSIAN planview.

This eBook is divided into two parts

Part One – Executive Summary, explains why mastering Lean and Agile principles and practices is necessary to improve an organization's ability to deliver customer-centric value and enhance business processes efficiently.

Part Two – Practicing SAFe, explores how SAFe's five disciplines of a Lean-Agile Organization provide the foundation for scaling Lean and Agile, solving complex business problems, and aligning strategy and execution.

Why Lean and Agile?

> "Adopting Lean product development can double labor productivity through the entire system, cut time-to-market for new products in half, and enable a wider variety of products within product families to be offered at very modest additional cost."[2]

By combining Lean and Agile methods, organizations can streamline processes while staying responsive to customer needs, ultimately driving better outcomes and continuous improvement:

- **Lean** focuses on eliminating waste, optimizing processes, and maximizing value with minimal resources. It drives efficiency by continuously identifying and removing non-value-adding activities in all work areas, from product development to delivery.
- **Agile** emphasizes adaptability, collaboration, and delivering working solutions frequently. It helps teams respond to change quickly, ensuring that products meet evolving customer needs and that feedback loops drive continuous improvement.

Lean-Agile organizational design prioritizes adaptability by flattening hierarchies, nurturing cross-functional collaboration, and aligning teams around customer value. With this approach, organizations can move beyond outdated functional silos and embrace team structures and lightweight processes that support faster value flow and create a culture of innovation and relentless improvement.

Key benefits of Lean and Agile include:

1. **Enhanced Customer Satisfaction:** Lean-Agile methods ensure that customer and business needs are addressed early and continuously, leading to higher satisfaction.
2. **Adaptability to Change:** Lean and Agile embrace changing requirements, even late in development, allowing teams to turn change into a competitive advantage.

[1] Gartner, *Market Guide for Enterprise Agile Frameworks*, by Keith Mann et al.
[2] James P. Womack and Daniel T. Jones, *Lean Thinking: Banish Waste and Create Wealth in Your Corporation* (New York: Simon & Schuster)

3. **Frequent Delivery of Working Software:** Lean-Agile emphasizes delivering innovative solutions frequently, with shorter cycles, ensuring continuous progress and timely feedback.

4. **Sustainable Development Practices:** Lean-Agile methods support sustainable, consistent development that reduces burnout, enhances collaboration, and maintains a steady pace.

5. **Continuous Improvement and Innovation:** Lean-Agile fosters a culture of technical excellence, innovation, and adaptability, ensuring continuous learning and improvement.

Having discussed the benefits of Lean and Agile, let's explore the core principles of these methodologies to understand their value and application better.

What is Lean and Agile?

> "Lean focuses on delivering maximum value with minimal waste, while Agile ensures adaptability and responsiveness to change. Together, they create a system that continuously improves, learns, and delivers customer value efficiently."
>
> —**Rupp and Knaster et al.,** paraphrased from *The Lean-Agile Way*[3]

The following two sections describe the key elements that form the basis of the Lean-Agile mindset shown in Figure 1. below.

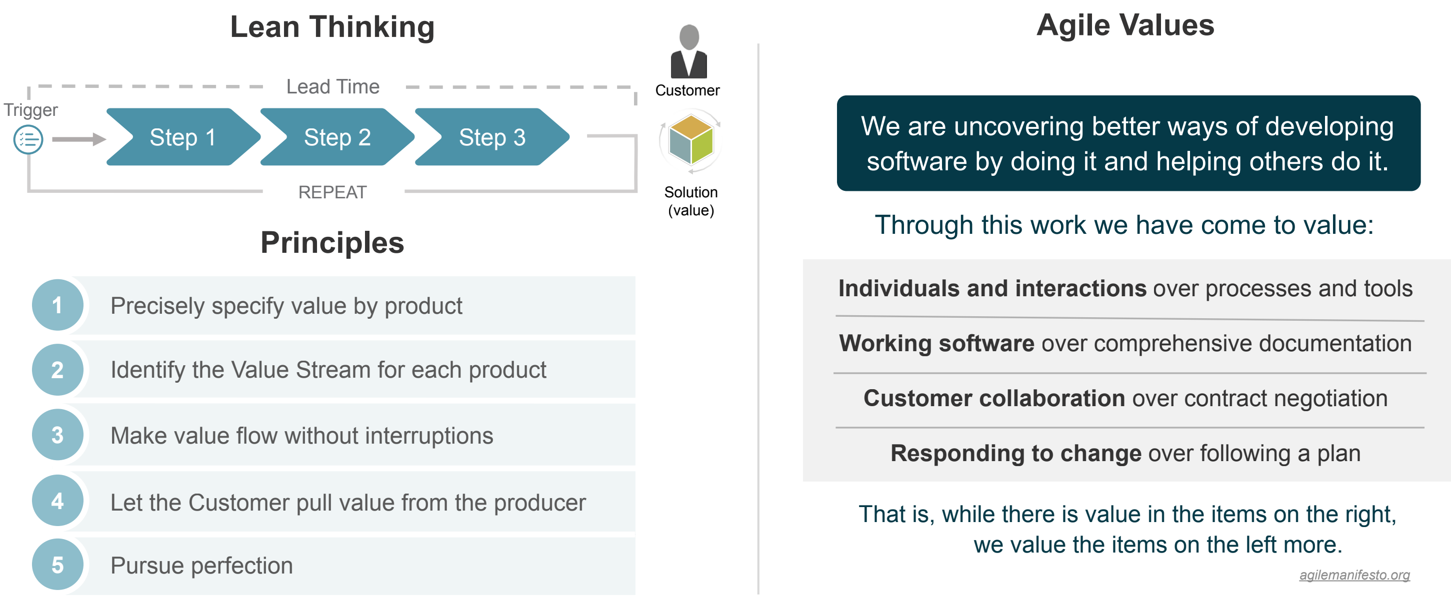

Figure 1. Lean Thinking and Agile Values

[3] C. Rupp & R. Knaster, et al. *The Lean-Agile Way*. Packt

What is Lean?

Initially inspired by Lean manufacturing, the principles and practices of Lean thinking have been extensively adapted for product development, offering a deep and comprehensive approach, as shown in Figure 2.

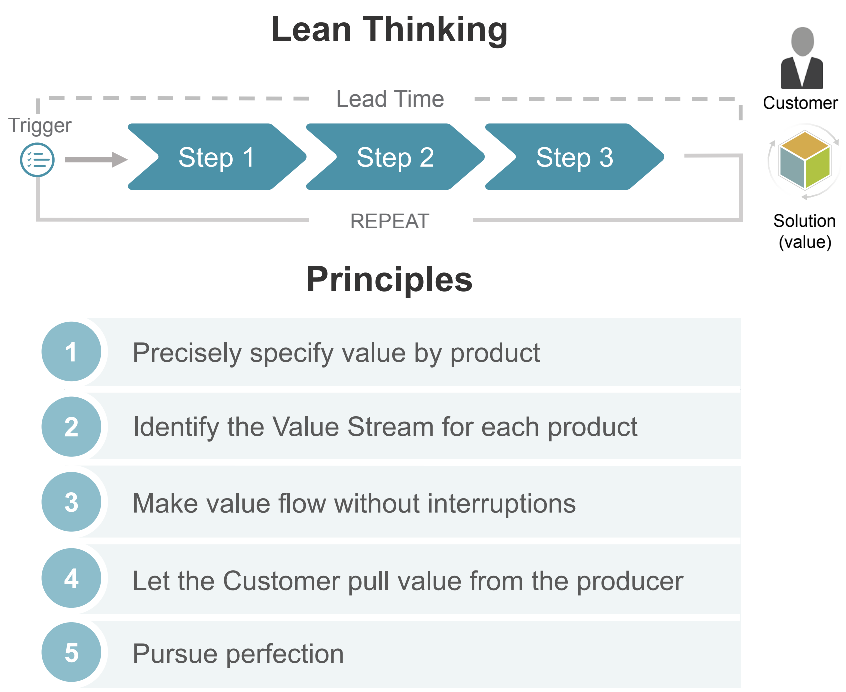

Figure 2. The Principles of Lean Thinking

- **The Goal of Lean Thinking** – Lean Thinking aims to deliver maximum value to the customer in the shortest sustainable lead time, from request to when the customer receives the value. How value is created also matters. High-quality, respect for people and society, high morale, safety, and customer delight are also essential goals and benefits of Lean Thinking. Achieving these goals requires applying the five basic principles of Lean Thinking illustrated in Figure 2, and described in the following paragraphs.
- **Precisely specify value by specific product** – Every organization is built to deliver value. This value is meaningful only when tied to a specific product that satisfies the customer's needs at a particular price and time. The first principle of Lean Thinking highlights the importance of understanding customers' needs and quantifying the value inherent in the solutions delivered to them. The product itself holds the value—not the project, initiative, or process that produces it—and only the ultimate customer can define that value.
- **Identify the Value Stream for each product** – Once the value is defined for each product and customer type, Lean Thinking describes how an organization creates that value, from identifying a need or opportunity to delivering the solution. This flow is the value stream and contains all the people, processes, tools, and information necessary to provide value. Delays anywhere in this system result in delayed delivery of value to customers.

- **Make value flow without interruptions** – Establishing an uninterrupted flow of work is critical for supporting incremental value delivery based on constant feedback and adjustment. Achieving value flow requires applying and understanding the eight fundamental properties of flow: visualizing and limiting work-in-process (WIP), addressing bottlenecks, minimizing handoffs and dependencies, getting fast feedback, working in small batches, managing queue lengths, optimizing 'time in the zone,' and remediating legacy policies and practices.
- **Let the customer pull value from the producer** – Organizations design value streams to deliver products based on customer demand—rather than pushing products into the market based on speculation—to achieve better alignment with market needs. This approach is essential for adjusting value stream capacity. Excess capacity leads to waste and inefficiency. Conversely, insufficient capacity creates bottlenecks and delays, disrupting the smooth flow of value to customers. The ideal state is to match capacity closely with market pull.
- **Pursue perfection** – The fifth and final principle of Lean—pursuing perfection—underscores that optimizing value flow is a continuous journey, not a destination. Even when the first five principles are diligently applied, market dynamics, evolving customer needs, and emerging technologies necessitate continuous refinement. By identifying what customers value or the problems they must solve, Lean ensures that every effort is directed toward delivering meaningful, high-impact products with fewer resources and people.

What is Agile?

> "Agile is an attitude, not a technique with boundaries. An attitude has no boundaries, so we wouldn't ask 'can I use Agile here,' but rather 'how would I act in the Agile way here,' or 'How Agile can we be here?"[4]
>
> —**Alistair Cockburn**, *Agile Software Development*

Alistair Cockburn reminds us that Agile is a mindset rather than a rigid set of rules, techniques, and defined roles. Instead of questioning whether Agile applies to a specific project, it should focus on how its principles and values can maximize effectiveness in any context. Agile emphasizes collaborating closely with customers rather than relying on contracts that often prevent change and unnecessary features. Teams actively engage with customers to get faster feedback and make continuous adjustments based on user input, ensuring the product evolves effectively.

An iterative development approach enables teams to deliver small increments of value frequently. This approach allows functional teams to play a crucial role in Agile, bringing together diverse skills needed to provide a product increment. These teams work collaboratively toward a shared goal, advancing efficiency and innovation. The values shown in Figure 3. illustrate the essence of Agile.

[4] Cockburn, Alistair. *Agile Software Development: The Cooperative Game*. Addison-Wesley Professional.

Agile Values

We are uncovering better ways of developing software by doing it and helping others do it.

Through this work we have come to value:

Individuals and interactions over processes and tools

Working software over comprehensive documentation

Customer collaboration over contract negotiation

Responding to change over following a plan

That is, while there is value in the items on the right, we value the items on the left more.

agilemanifesto.org

Figure 3. Manifesto for Agile Software Development

- **Individuals and Interactions** Over Processes and Tools – Agile frameworks like Scrum, Kanban, and SAFe matter, but processes should serve people, not the other way around. When a process fails, adapt it. Tools support communication but shouldn't replace human interaction.
- **Working Software** Over Comprehensive Documentation – Documentation has value but should evolve with Lean-Agile working methods. Prioritize delivering tangible work products for customer feedback over excessive early documentation.
- **Customer Collaboration** Over Contract Negotiation – Customers define value, making ongoing collaboration crucial. Contracts set expectations but shouldn't hinder flexibility or trust. Favor win-win agreements that foster long-term partnerships.
- **Responding to Change** Over Following a Plan – Change drives agility. Plans should adapt as understanding, stakeholder insights, learning, and customer needs evolve. Measuring success by rigid adherence to a plan leads to inefficiency—embrace change instead.

Empowered teams are a cornerstone of Agile, as they are encouraged to be self-managing and make decisions independently. This autonomy nurtures creativity, accountability, and a strong sense of ownership among team members. Another key aspect of Agile is to embrace change, allowing teams to adapt to new information and evolving requirements. Regular planning and review cycles ensure alignment and responsiveness, keeping the process dynamic and customer-focused.

There is overwhelming evidence from success stories in all industries across every geography demonstrating the extraordinary business and personal benefits of a Lean and Agile way of thinking and working. The following section explores how SAFe applies Lean-Agile principles to empower teams at scale, driving alignment, fostering adaptability, and enabling continuous value delivery across the organization.

Why SAFe?

> ## Work Differently. **Build the Future.**
>
> —Scaled Agile Inc. mission statement

Adopting a proven framework like SAFe accelerates Lean and Agile adoption by providing structured guidance. This enables organizations to focus on delivering customer value rather than designing new processes.

Initially, Agile was designed for small, cross-functional, co-located teams that could collaborate closely and iterate quickly. However, implementing Agile successfully in a large organization presents a whole new set of challenges.

These challenges significantly increase when scaling to multiple teams and building complex products that may involve hundreds or even thousands of people, including external suppliers and partners. Key concerns include:

- Aligning strategy and execution and prioritizing initiatives
- Reducing risk through incremental funding and development
- Coordinating work and dependencies across multiple teams
- Synchronizing planning, development, and releases
- Scaling Agile beyond IT to the entire organization

SAFe exists to help organizations striving for excellence build winning products by providing a knowledge base of the most effective, proven product development practices. It offers an operating model for implementing Lean-Agile practices across large organizations. SAFe guides how businesses and employees can grow their knowledge and skills and learn to sense and respond to change faster. Moreover, SAFe is designed to help organizations scale Agile practices to larger initiatives with multiple teams. Instead of having each team work independently and coordinate their efforts at the end, it encourages all teams to work together from the start, sharing information and collaborating on the same objectives.

The SAFe website[5] offers an interactive Big Picture (BP) graphic — which provides an overview of the Framework and helps you navigate the knowledge base as shown in Figure 4—each icon on the BP links to SAFe's guidance on that subject.

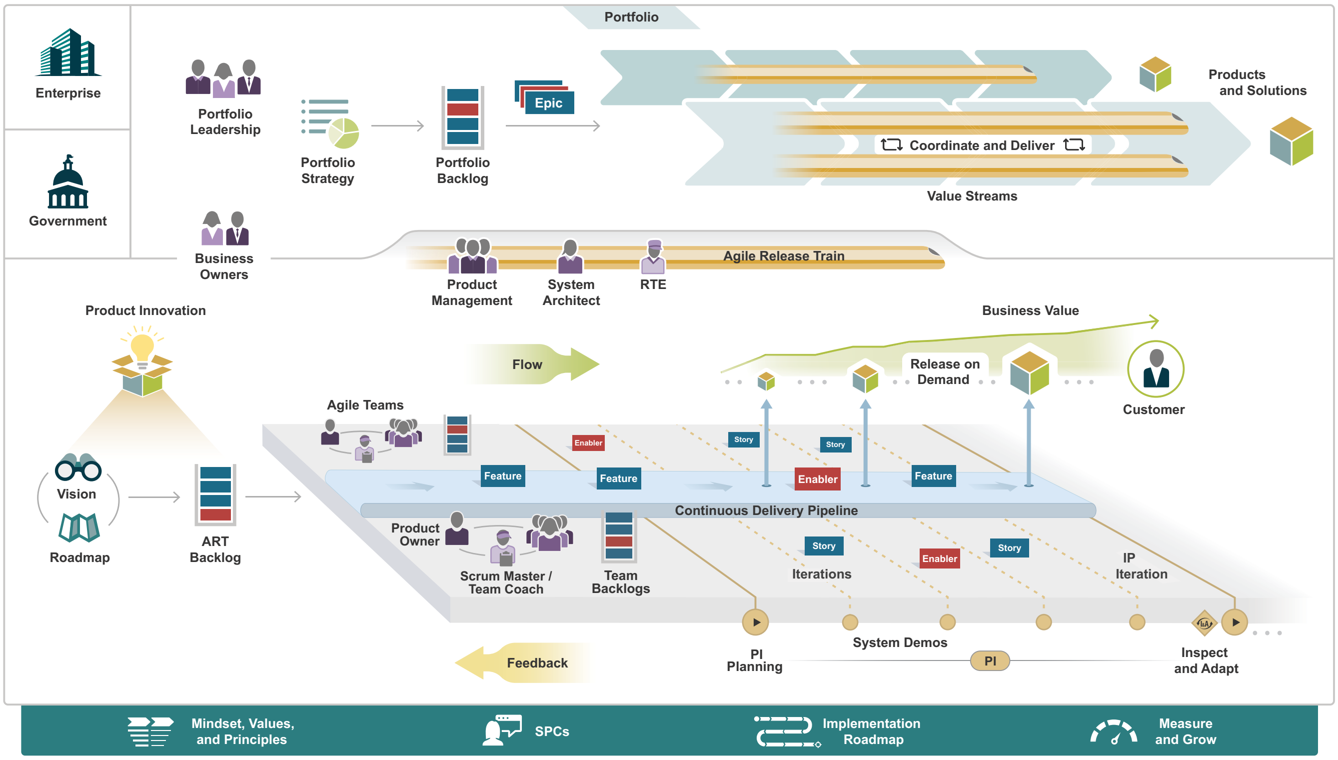

Figure 4. The SAFe Big Picture

The SAFe Advantage: Integrating Lean and Agile at Scale

Structuring Agile teams and teams of Agile Teams (Agile Release Trains) around specific products and aligning them with common goals and objectives, **SAFe maximizes customer value while minimizing waste.** This integration enables organizations to respond swiftly and effectively to market changes, ultimately enhancing customer satisfaction and business outcomes. It boosts productivity and cultivates a culture of relentless improvement and innovation within the organization.

A key aspect of SAFe is establishing a regular rhythm for planning and execution. Synchronizing the efforts of multiple teams helps **maintain alignment with the overall business strategy,** enabling work to progress in a flexible, coordinated, and predictable manner. This approach fosters collaboration and accelerates delivery.

Both Lean and Agile emphasize the importance of measurement and feedback. **SAFe leverages key metrics to track progress, quality, and performance,** enabling data-driven decisions and continuous process refinement. Regular feedback loops empower teams to adapt quickly and improve their outcomes.

Successfully adopting Lean-Agile practices in SAFe relies on a strong leadership culture. **Leaders play a critical role** in creating an environment that supports experimentation, ongoing learning, and cross-team cooperation, ultimately driving sustainable success.

[5] https://framework.scaledagile.com

What is SAFe?

The guidance in the SAFe Framework is organized around five disciplines, as shown in Figure 5. Each discipline forms a critical component of a Lean-Agile organization. The disciplines describe the knowledge, skills, and techniques required to achieve mastery. They provide the necessary information, learning resources, and practical application guidance to support success.

Figure 5. Five Disciplines of SAFe

The following sections briefly describe each of the five disciplines. In Part 2, we will explore these disciplines in further detail.

Leadership and Culture

> "Leaders must wire their organizations to create conditions where people can solve problems well and systematize new solutions. By creating and sustaining good social circuitry, individual contributions can combine into a collective effort toward a common purpose. It is the leader's responsibility to ensure people are able to use their energy and time in ways that are productive, appreciated, and value-adding."
>
> —**Gene Kim**, *Wiring the Winning Organization*

Leaders model accepted organizational behaviors through their words and actions. How leaders show up sets the tone for the organization's culture, for better or worse. The most effective technique for driving cultural change is for leaders to demonstrate the behaviors and mindsets needed to succeed so that others can learn and grow by their example.

The Leadership and Culture Discipline (L&C) describes how leaders can create a positive, performance-oriented culture that enables individuals and teams to reach their highest potential. Implementing this culture ensures that the 'social circuitry' that is foundational to the success of adopting SAFe as an organization's way of working.

Team and Technical Agility

> "Continuous attention to technical excellence and good design enhances agility."
>
> —**The Agile Manifesto**

The Team and Technical Agility (TTA) discipline describes how to enable cross-functional Agile teams to accelerate value delivery. It describes the critical skills, roles, and practices that high-performing Agile Teams and Agile Release Trains (ARTs) use to create high-quality products and services for customers.

All Agile Teams and ARTs are responsible for building quality into everything they deliver. Applying this discipline to all product development, engineering, and delivery results in an organization with the team and technical agility required to provide superior value at scale.

Product Development Flow

> "Building a successful product is not about delivering features—it's about creating value for users and the business."
>
> —**Roman Pichler**, Author, Renowned Product Management Expert

The Product Development Flow (PDF) discipline enables organizations to smoothly release valuable product increments and respond swiftly to market changes. By encouraging consistent feedback and ongoing improvement, companies can drive innovation and deliver value to customers. Product innovation is key to adapting to emerging market opportunities. They maintain their competitiveness and relevance by focusing on value flow, continual delivery, and customer-centric practices. These practices require balancing execution and understanding customer needs—delivering the right products at the right time.

The traditional "build it, and they will come" approach no longer holds true for most organizations. Building a successful operating model around a product goes beyond development; it requires navigating the complexities of product-market fit, scaling through sales and marketing, and steady innovation to keep up with ever-shortening product life cycles. Lean-Agile ways of working enable organizations to rapidly deliver innovations that keep pace with market trends.

Large Solution Integration and Delivery

> "Building large software solutions is not just about writing code; it's about orchestrating a complex ecosystem of people, processes, and technology, all while embracing the constant flux of requirements and evolving landscapes."
>
> —**Mike Cohn**

The Large Solution Integration and Delivery (LSID) discipline applies SAFe principles to developing, evolving, and operating the world's largest systems. LSID solutions typically require the work of many value streams and ARTs that need to align on delivery.

These large solutions often involve hundreds or even thousands of engineers and must usually meet strict regulatory and compliance requirements. The inherent complexity not only encompasses technical challenges but also consists of crafting user journeys that span multiple products and business lines.

Lean Portfolio Management

> "Most strategy dialogues end up with executives talking at cross-purposes because … nobody knows exactly what is meant by vision and strategy, and no two people ever quite agree on which topics belong where. That is why, when you ask members of an executive team to describe and explain the corporate strategy, you frequently get wildly different answers."
>
> —**Geoffrey Moore**, *Escape Velocity*

Traditional approaches to portfolio management were not designed to compete in today's fast-paced economic environment. Organizations face more uncertainty and pressure to deliver innovative products faster. Many legacy portfolio practices remain despite massive market changes. Modernizing portfolio management is critical to adopting a Lean-Agile way of working and achieving the strategic agility required to compete.

The Lean Portfolio Management (LPM) discipline aligns strategy and execution by applying Lean thinking to strategy and investment funding, Agile portfolio operations, and governance.

Benefits of SAFe

> "Our research tells us that implementing SAFe can deliver significant, measurable business value in complex initiatives by coordinating the work of multiple teams."
>
> —**Gartner Research**: *10 Essential Practices for Success When Implementing SAFe*

SAFe's proven practices drive organizational agility, aligning the whole organization to the business strategy, fostering innovative products that delight customers, improving the speed of value delivery, and reducing waste and delays. Every business needs the agility to thrive and survive. Rapid execution, exceptional quality, and exceeding customer expectations are essential for all organizations. Success relies on delivering high-quality, innovative products and services quickly and sustainably.

> **Commercial Banking finds that SAFe supports multiple teams in building large solutions.**
> "The products we're developing are bigger than one Agile team. For the teams to interact and plan together, we really needed SAFe as the foundation. It brings the practices and methodologies to coordinate multiple teams working on the same product at the same time."
>
> —**Mike Eason**, Capital One CIO, Commercial Banking

SAFe benefits all levels of business, from individual developers to the entire organization, as shown in Figure 6.

- Based on case studies, organizations adopting SAFe achieved a 50% increase in time to market and quality, a 35% increase in productivity, and a 30% increase in employee engagement.
- Large and complex organizations get a proven 'operating model' and guidance for creating working products in a better, more Agile, and leaner way.
- Teams can focus more intensely on the customer and create higher quality products faster and sustainably, even in a large enterprise.
- Developers find work more enjoyable by gaining autonomy and purpose, and reducing confusion and waste.

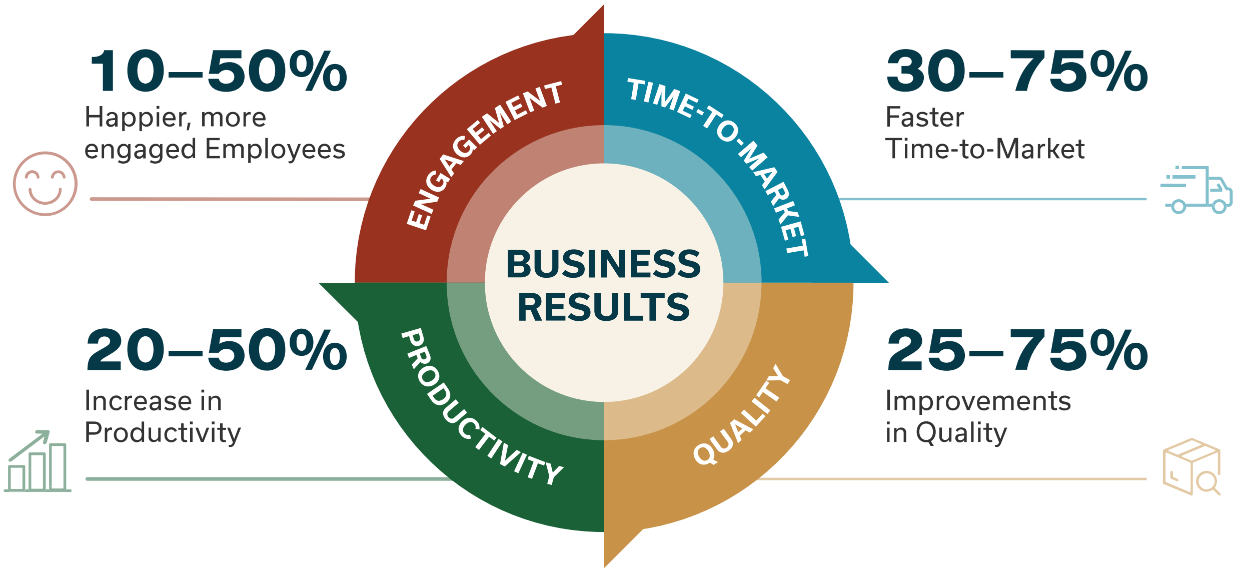

Figure 6. SAFe Business Results

In summary, SAFe helps teams work smarter, deliver faster, and improve quality while boosting morale. It allows teams to choose how they organize work around products and customer needs instead of rigid hierarchies. It encourages persistent teams to maintain knowledge and reduce burnout. By prioritizing built-in quality, SAFe prevents costly fixes later and reduces technical debt. It also simplifies planning by aligning teams with business goals, coordinating work across teams, and enabling direct collaboration with decision-makers.

Moreover, teams feel more engaged because they have a say in their work, receive quick feedback, and operate in a culture that values improvement and respect for people. Morale improves as SAFe encourages winning, collaboration, and meaningful contributions to business success. Additionally, it changes the nature of leadership by developing Lean-Agile Leaders who guide with trust and intent rather than rigid control, creating an environment where teams feel motivated and inspired.

Moving Beyond Social Media Hype: Agile is Dead

For several years, social media has frequently proclaimed that “Agile is dead,” a sensationalized and misguided claim. Agile continues to evolve and thrive, adapting to modern business challenges by integrating technologies like AI and machine learning and expanding beyond software development into diverse business functions. Its core principles—adaptability, collaboration, and customer focus—remain indispensable. The 2024 State of Agile Report[6] underscores its enduring value, showing increased satisfaction and adoption across industries—furthermore, only a few desire to return to traditional delivery methods. A superior alternative has not yet emerged to replace its proven success.

Unlocking the Potential of Agile

Teams must break free from outdated habits, challenge assumptions, and embrace meaningful change, as described here.

- **Break Free from Strict Compliance:** Agile is a mindset focused on adaptability, not a rigid set of rules. Evaluate whether Agile practices add value in your context and adjust as necessary. Agile should empower teams, promoting open discussion and customization to meet unique needs.
- **Go Beyond the Buzzwords:** Commit to genuine Agile adoption rather than fake implementation. Assess if your team’s practices are genuinely Agile or outdated methods in disguise. Focus on working smarter and fostering a culture of transparency, experimentation, and continuous improvement to drive real change and innovation.

Overcoming Agile Challenges

The biggest obstacle to agility isn’t a lack of process—it’s the unwillingness to question, experiment, adapt, and evolve. Too often, teams get stuck in legacy practices, preventing progress. You’re not alone in this journey—Scaled Agile’s network of over 400 partners in 50+ countries is available to train your teams and support SAFe adoption, as shown in Figure 7.

Additionally, 2 million SAFe-trained professionals in 110+ countries demonstrate a vast, global community of expertise and support. Many trained professionals indicate widespread industry recognition and experience, ensuring organizations can find skilled practitioners and trainers to guide their SAFe adoption.

[6] www.stateofagile.com

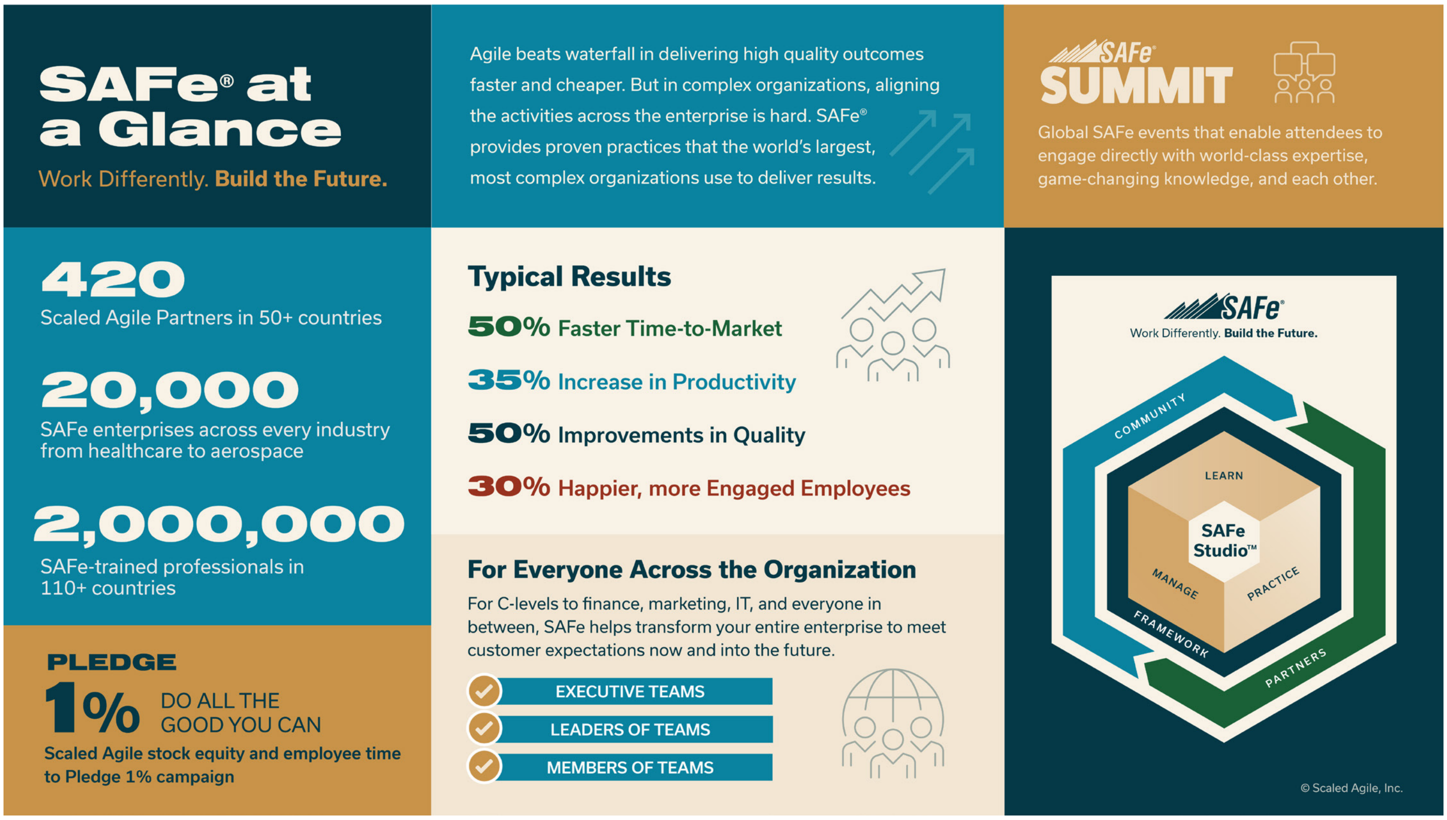

Figure 7. SAFe by the Numbers

Getting Started: A Proven Method for Scaling Change

> "Any successful change requires a translation of ambiguous goals into concrete behaviors. To make a switch, you need to script the critical moves."
>
> —**Chip Heath and Dan Heath**, *Switch: How to Change Things When Change Is Hard*

Organizational transformations have a notoriously high failure rate, with studies indicating that up to 70% of large-scale change initiatives fail to achieve their desired objectives.[7] This high failure rate can be attributed to several factors, including:

- There is no clear and compelling business case for change
- Poor alignment among and engagement from leadership
- Failure to set goals, create urgency, or clarify the "why"
- Slow decision-making and inaction on problems
- Pursuing large-scale change instead of an iterative approach

[7] https://www.mckinsey.com/capabilities/transformation/our-insights/perspectives-on-transformation

Given these challenges, a proven SAFe implementation roadmap dramatically reduces the risk of failure. The roadmap provides a step-by-step guide with 13 critical moves based on the experiences of hundreds of organizations. While each SAFe adoption is unique, those that achieve the best results follow a similar path to that outlined in the Implementation Roadmap. However, success relies on customizing it for your context. Each organization's needs and journey will differ based on agile maturity and tolerance for change.

A critical question leaders often ask is, where do I begin implementation? While successful change initiatives are often driven from the top down, they frequently originate at the bottom or in the middle. Agile movements typically start within teams. Executives' pay attention when teams succeed.

The answer as to where to start depends on your organization's context and goals. Below are some simple guidelines:

- **New to Agile or gradual adoption:** Start with Agile teams and ARTs to build a strong foundation in Team and Technical Agility, enabling quick scaling.
- **Need rapid market response:** Focus on Product Development flow to drive innovation, early delivery, and customer-centric practices for faster time to market.
- **Organizations facing alignment challenges:** Start with Lean Portfolio Management to connect strategy with execution, set strategic direction, prioritize large initiatives, and allocate people and resources based on business needs.

There are several other starting points for your journey, and any of them will propel your organization toward superior business outcomes. You should work with an experienced SAFe partner to chart your path based on your organization's goals and context. While the SAFe roadmap provides a structured approach, adapting it to align with your company's goals and needs is essential.

PART TWO

Practicing SAFe

Practicing SAFe involves applying it to real-world scenarios to drive agility, alignment, and value delivery across teams and the organization. It requires a combination of mindset shifts, process adoption, and learning to ensure successful ongoing change and improvement. Organizations must promote collaboration, embrace Lean-Agile leadership, and leverage SAFe to improve execution, enhance innovation, and respond effectively to changing market demands.

The following sections further describe SAFe's five disciplines introduced in Part 1. These disciplines enable organizations to achieve customer-centricity and better business outcomes.

Leadership and Culture

An organization's leaders are responsible for adopting and improving Lean-Agile development and the disciplines and competencies that enable this change. Only they have the authority to change and improve the systems that determine how work is done. Moreover, only these leaders can create an environment that encourages high-performing Agile teams to flourish and produce value. Leaders represent accepted organizational behaviors through their words and actions. How leaders do this sets the tone for the organization's culture. The most effective technique for driving cultural change is for leaders to demonstrate the behaviors and mindsets needed to succeed so others can follow.

The Leadership and Culture (L&C) discipline, illustrated in Figure 8, describes how leaders can create a positive, generative culture (performance-oriented) that enables individuals and teams to reach their highest potential. This discipline and its supporting competencies form the foundation for successfully adopting SAFe.

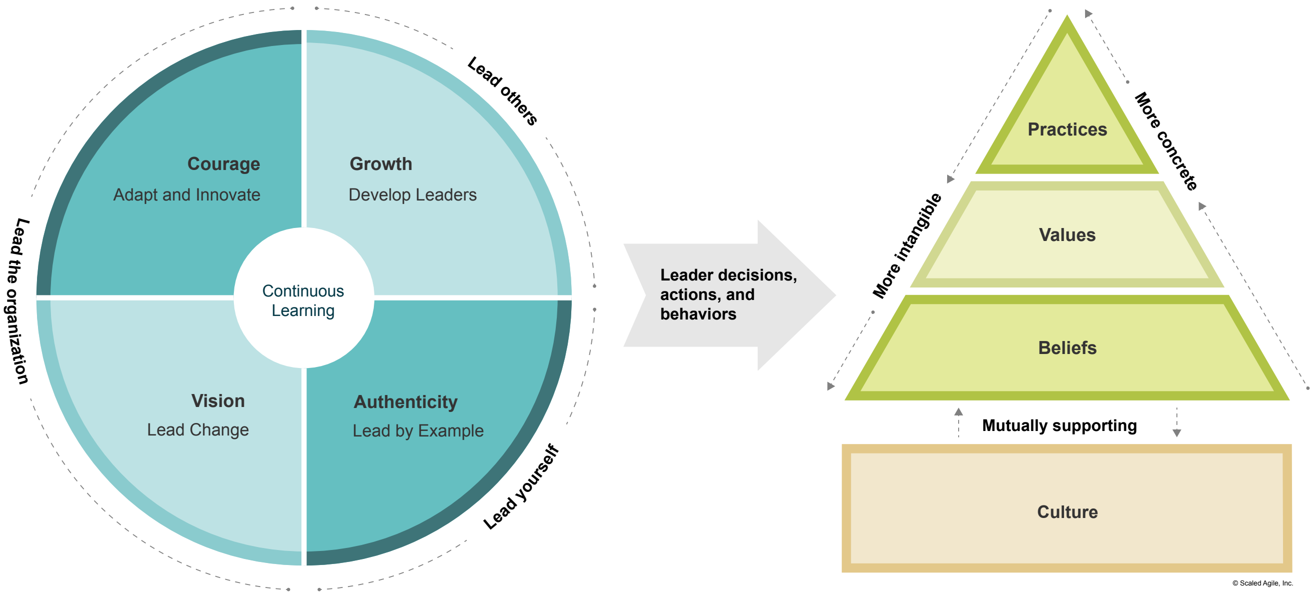

Figure 8. Leadership and Culture Discipline

The following sections provide a more detailed overview of the elements of the Leadership and Culture discipline.

Leadership

What actions and behaviors should leaders embrace to set the right example and build a culture that supports the Lean-Agile working methods? The behaviors illustrated in Figure 8 are inspired by a modern leadership style and based on four fundamental behavior patterns:

1. **Leading By Example:** Leadership is most effective when actions align with words. Credibility comes from authenticity, consistency, and continuous growth, not just authority.
2. **Growing Leaders:** Starts with trust-based leadership that empowers others through clear intent, not micromanagement. Supporting growth with coaching ensures people feel valued and capable.
3. **Leading Change:** Change must be inspired, not mandated. Effective leaders provide vision, mobilize teams, and swiftly remove obstacles.
4. **Adapting and Innovating:** Leaders set the tone by creating an environment where people are empowered to challenge the status quo, experiment with new ideas, and improve continually. By embracing adaptability, they build resilience to change and a forward-looking organization.

These leadership behaviors stem from vision, authenticity, growth, and courage. Leaders who embody these traits guide themselves, their teams, and organizations. Leadership is an ongoing journey of growth and learning.

Culture

Leaders shape culture through their actions, setting patterns that define how work gets done. The most effective way to drive cultural change is by modeling the desired behaviors.

A generative culture, as defined by Dr. Ron Westrum, fosters teamwork, open communication, and a shared focus on goals. Trust, transparency, and a learning mindset drive collaboration and continuous improvement.

Cultural change is gradual and occurs only when a new way of operating has succeeded over a minimum amount of time. It follows demonstrated success, not precedes it. Leaders must reinforce desired behaviors through recognition, decisions, and structural changes that support collaboration. By aligning with generative culture principles, organizations create an environment where SAFe adoption, agility, and innovation can thrive.

Team and Technical Agility

The Team and Technical Agility (TTA) discipline describes how to enable cross-functional Agile teams to accelerate value delivery. It describes the critical skills, roles, and practices that high-performing teams and Agile Release Trains (ARTs) use to create high-quality customer products. Teams are responsible for building quality into everything they deliver. Inspection often comes too late to be effective. Applying the TTA discipline to all product development, engineering, and delivery results in an organization that can provide value at scale.

Figure 9. illustrates the elements, processes, and outcomes of Team and Technical Agility.

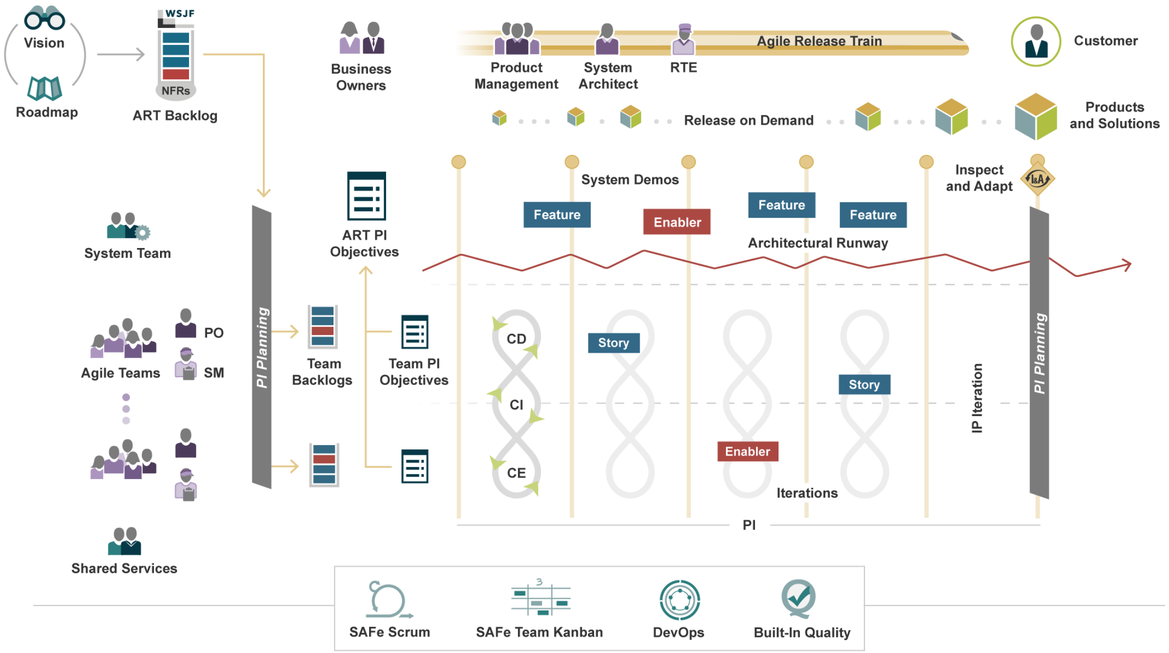

Figure 9. Team and Technical Agility Discipline

Agile Release Trains (ARTs)

When a product is too large for a single team to deliver value, an ART effectively aligns multiple teams to a shared vision and roadmap.

ARTs help teams work together smoothly and ensure that the value they are developing is ready for use simultaneously. ARTs usually have 50–125 people. They are cross-functional and have all the capabilities needed to define, build, validate, release, and, where applicable, operate one or more products.

The ART operates with three critical leadership roles that fulfill the responsibilities of content authority, technical authority, and value delivery:

1. **Product Management** – Responsible for defining and supporting the building of desirable, feasible, viable, and sustainable products that meet customer needs.

2. **System Architect** – An individual or small team that defines the system's overall architecture, helps identify Nonfunctional Requirements (NFRs), determines the significant elements and subsystems, and helps design the interfaces.

3. **Release Train Engineer (RTE)** – A servant leader and ART coach who actively facilitates ART events and processes while empowering and supporting teams to deliver value consistently.

Business Owners are the primary stakeholders of the ART and actively participate in their events. They have the business and technical responsibility for fitness for use, governance, and return on investment (ROI) for products.

ART Events

The heartbeat of the ART is Planning Intervals (PIs), which represent fixed timeboxes during which the ART steadily delivers value to its customers. PIs are typically 8-12 weeks and provide a development rhythm for ARTs. This includes one iteration focused on Innovation and Planning (IP).

ART events are critical for ensuring alignment, collaboration, and predictable progress across multiple Agile teams working toward common goals. They provide a regular rhythm for planning, reviewing, and coordinating efforts, helping teams stay on track and adapt to changes as needed.

Below are the three key events that guide an ART's operations.

1. **PI Planning:** A regularly scheduled event that brings all teams on the ART together to align to a shared mission and create a plan for delivering a set of committed PI objectives.

2. **System Demo:** A demonstration at the end of each iteration or PI, showcasing the integrated work from all teams, enabling stakeholders to assess progress and provide feedback.

3. **Inspect and Adapt (I&A):** An event at the end of the PI that enables teams to evaluate the current product, reflect on their processes, and identify areas for improvement through a structured problem-solving workshop.

Regular Coach and PO Sync events also streamline communication and coordination across the ART. The Innovation and Planning (IP) Iteration allows teams more time for exploration, innovation, and dedicated planning. This time can be used for informal and formal learning opportunities, fostering a spirit of growth and improvement.

Agile Teams

Agile Teams are cross-functional groups of ten or fewer individuals who can define, build, test, and deploy an increment of value in a short time box. To facilitate their work, most teams adopt SAFe Scrum, a lightweight process designed to deliver value rapidly. Likewise, SAFe Team Kanban—also a Lean-Agile method—helps teams visualize their workflows, establish Work in Process (WIP) limits, and measure throughput. Both approaches emphasize the importance of delivering value continuously while encouraging ongoing process improvement.

Each Agile Team typically includes the following three roles that fulfill the responsibilities of content authority, technical authority, and value delivery.

1. **Product Owner (PO):** The content authority for the team backlog, responsible for defining and accepting user stories, prioritizing work, and ensuring the team delivers value aligned with customer and business goals.

2. **Agile Team Members:** Individuals who collaborate to define, build, test, and release valuable increments. Each team member contributes to delivering high-quality products through shared responsibility and expertise.

3. **Scrum Master (SM):** A servant leader and Agile coach who removes obstacles, facilitates team events, and promotes a high-performance, self-managing team environment.

Team Events

Team events are essential for Agile Teams to plan, review, and reflect on their work. These activities ensure alignment, improvement, and effective value delivery throughout the iteration, which is typically 2 weeks.

Below are the three key events that enable Agile Teams to structure their process.

1. **Iteration Planning:** The team selects backlog items they can commit to and defines iteration goals.

2. **Iteration Review:** The team inspects the increment, evaluates progress, and updates the backlog.

3. **Iteration Retrospective:** The team reflects on the iteration, assesses practices, and identifies improvements.

Additionally, teams hold sync events that coordinate their daily activities and address blocking issues and dependencies.

Technical Agility Practices

The ART builds or shares a Continuous Delivery Pipeline (CDP), allowing teams to deliver new functionality to users as needed. In some instances, this may mean daily or even multiple releases per day. In other organizations, releasing is weekly or monthly to satisfy market demands and business goals. To support frequent value delivery, the ART builds an Architectural Runway consisting of the existing code, components, and technical infrastructure needed to implement near-term features with minimal redesign and delay.

A DevOps mindset fosters communication, integration, and automation across all system development and operations stages to enhance collaboration and efficiency. Additionally, built-in quality is critical to scaling agility across multiple teams. This approach describes a set of practices to ensure that teams' outputs consistently meet appropriate quality standards while creating customer value.

Product Development Flow

The Product Development Flow (PDF) discipline enables organizations to release valuable product increments and respond swiftly to market changes. Companies can drive innovation and deliver customer value by encouraging regular feedback and improvement. Product innovation is key to adapting to emerging market opportunities. These organizations maintain their competitiveness and relevance by focusing on the flow of value, constant delivery, and customer-centric practices.

Building a successful operating model around a product goes well beyond development. It requires achieving product-market fit, scaling through sales and marketing, and steady innovation to adapt to shorter product life cycles. Lean-Agile methods help organizations rapidly deliver innovations that align with market trends.

Figure 10. shows the elements, processes, and outcomes of product development flow.

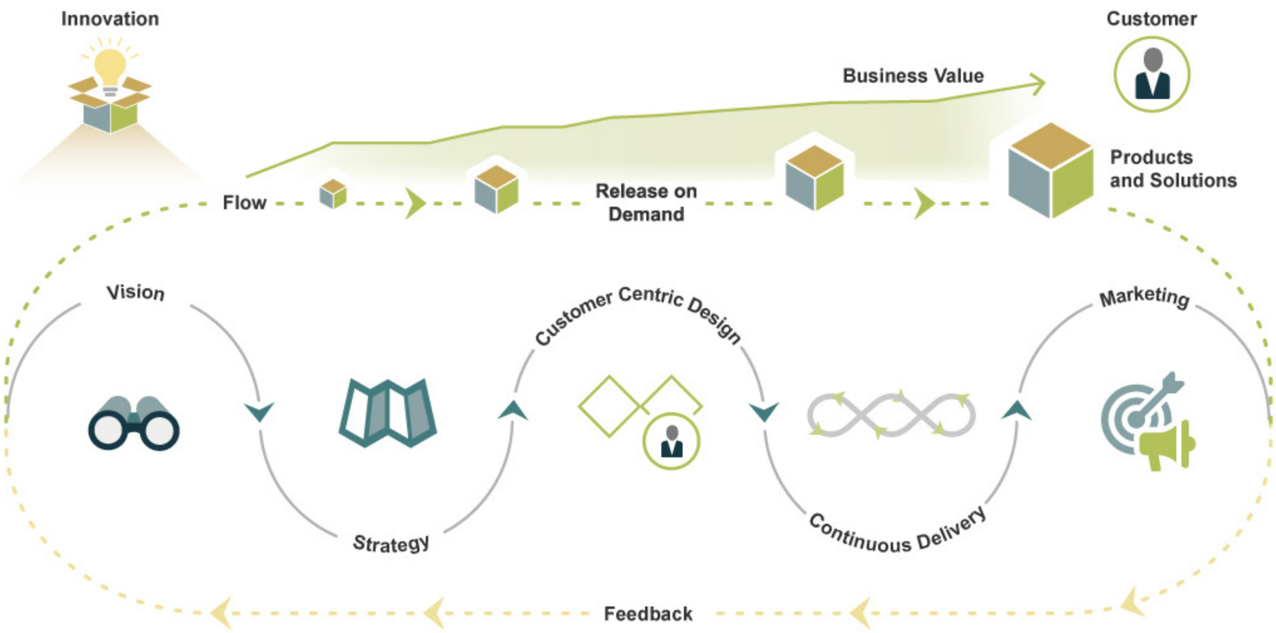

Figure 10. Product Development Flow Discipline

To achieve product development flow, organizations need to adopt the following concepts and practices:

- **Vision:** A clearly defined product vision intended for the future state that aligns teams and drives customer success.
- **Strategy:** A flexible product roadmap provides a clear path for achieving the product vision while constantly adapting to market changes.
- **Customer-Centric Design:** A Design Thinking approach that prioritizes the customer experience and problem-solving.
- **Continuous Delivery:** A streamlined process that ensures efficient and timely movement of ideas from concept to market.

- **Product Marketing:** Strategic communication that highlights the product's value and resonates with the target audience.
- **Feedback systems:** Rapid customer feedback and product data encourage collaboration and the generation of innovative ideas and solutions.

Rapid delivery of business value is a key driver of customer satisfaction and loyalty, increasing the likelihood of repeat customers. Organizations that consistently provide timely products gain a significant competitive advantage, as customers naturally gravitate towards products and services that swiftly address their needs.

Accelerating product flow nurtures a learning culture where ongoing improvement and innovation are prioritized. Agile teams should establish fast feedback loops to gather insights from customers and markets, refining their offerings accordingly. The team's processes should promote new ideas, experimentation, and rapid iteration to create an environment where creativity thrives.

Large Solution Integration and Delivery

The Large Solution Integration and Delivery (LSID) discipline applies SAFe to developing, operating, and evolving complex software, hardware, and cyber-physical systems.

The LSID discipline is focused on solutions that may require the work of many people in different value streams and ARTs, which need to align on delivery. For simplicity's sake, the various scaling scenarios are collectively described as 'large solutions,' regardless of whether six ARTs are building an organization-level software application or 60 ARTs are delivering a cyber-physical solution such as a defense system, satellite, or autonomous vehicle.

Figure 11. shows the elements, processes, and outcomes of the Large Solution Integration and Delivery discipline. This discipline scales up the practices described in the Team and Technical Agility discipline, which covers the more common scenario of Agile Teams working as part of a single ART across software and hardware.

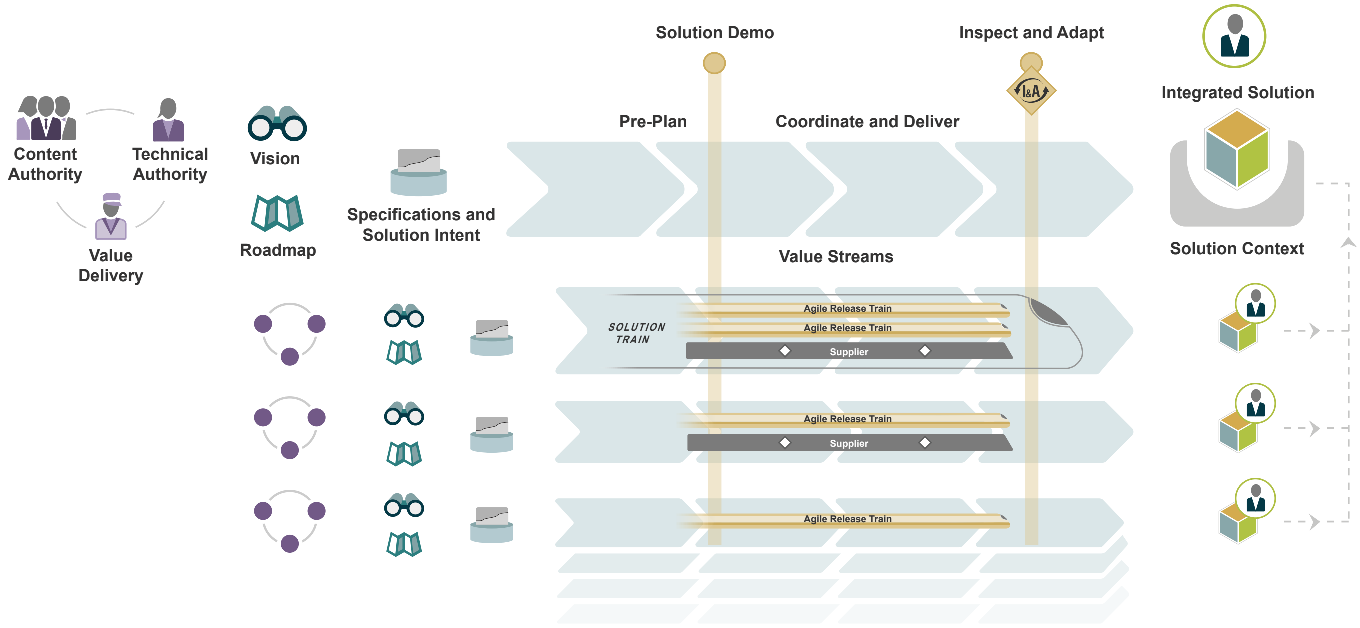

Figure 11. Large Solution Integration and Delivery Discipline

As shown in Figure 11, building and evolving large solutions is a significant undertaking that requires careful planning and execution. These large solutions often involve hundreds or even thousands of engineers and must usually meet strict regulatory and compliance requirements. The inherent complexity not only encompasses technical challenges but also consists of crafting user journeys that span multiple products and business lines. In the case of cyber-physical systems, a diverse range of engineering disciplines is necessary, along with the integration of hardware with long lead-time components. Given these challenges, it's crucial to implement sophisticated and rigorous practices in engineering, operations, and system evolution.

Large solutions emerge from large value streams comprising multiple Solution Trains, ARTs, teams, and suppliers. Value streams can be nested under a parent value stream providing a unified vision, roadmap, and solution intent to scale further and create alignment. A shared PI cadence ensures synchronization across planning, alignment, review, and improvement activities, fostering cohesion and efficiency.

Each value stream serves a customer. The ultimate customer of the integrated solution—the end-user or operator—resides at the end of the parent value stream. Internal customers also exist at the end of each nested value stream. They are often the teams consuming the work of that value stream so they can then provide a larger, integrated unit of value.

Throughout SAFe, Solution Trains, ARTs, and Agile Teams each operate with three critical roles that fulfill the responsibilities of content authority, technical authority, and value delivery. For example, Product Management, System Architect, and Release Train Engineer on an Agile Release Train. When developing large solutions, these roles require even more intentional connection and clear communication. The representative authority, regardless of the role name, is shown in Figure 11. Together, they form the leadership that works together to assume responsibility for the success of building the large integrated solution.

Organizations that stick to traditional methods—like detailed upfront specifications and designs based on a speculative outcome—may struggle to keep up. They miss out on opportunities to innovate and create leading-edge solutions that leverage the collective intelligence of many solution developers.

Lean Portfolio Management

Traditional approaches to portfolio management were not designed to compete in today's modern age. Organizations today face more uncertainty and pressure to deliver innovative products faster. Many legacy portfolio practices remain despite massive market changes. Modernizing portfolio management is critical to adopting a holistic Lean-Agile way of working and achieving the strategy agility required to compete.

The Lean Portfolio Management (LPM) discipline aligns strategy and execution by applying Lean thinking to strategy and investment funding, Agile portfolio operations, and governance.

Figure 12. shows the elements, processes, and outcomes of the LPM discipline.

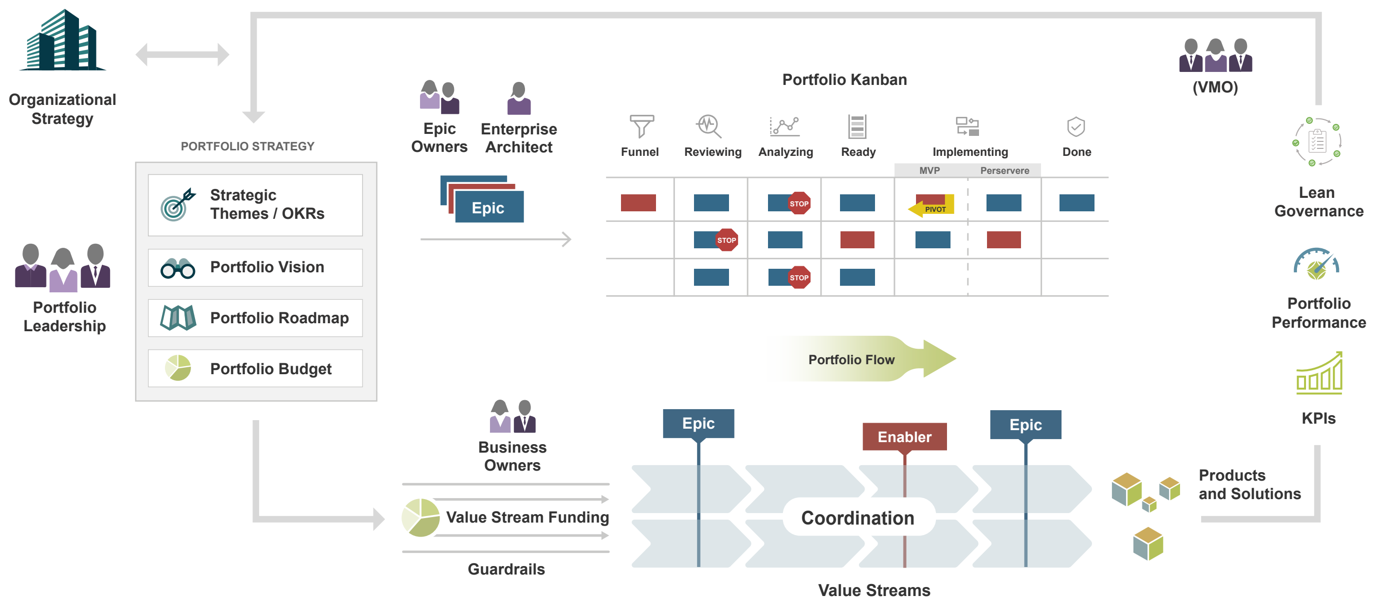

Figure 12. Lean Portfolio Management Discipline

As shown in Figure 12, Portfolio Leadership is responsible for formulating and communicating a portfolio strategy that aligns with the broader organizational strategy. They are also accountable for maintaining a portfolio vision and portfolio roadmap of strategic initiatives. The portfolio itself is organized around a collection of value streams that deliver a continuous flow of products and solutions to customers. The available portfolio budget is directly allocated to these value streams rather than the traditional approach of project-based funding. This ensures that each value stream has the flexibility to prioritize, ensuring they are always delivering the most important work.

Epics describe the strategic initiatives for which the portfolio is responsible. Epic Owners shepherd these Epics through the portfolio Kanban system, which brings transparency to the flow of work. The Lean Startup approach ensures that a Minimum Viable Product (MVP) is tested before a commitment to a full implementation. At any point, work items that don't move forward in the portfolio strategy are removed from the Kanban system. As the work to implement epics moves to the Agile Release Trains (ARTs), Business Owners on those ARTs ensure the work stays in alignment with the strategic intent.

The Value Management Office (VMO) is accountable for the portfolio's governance and day-to-day operations, including the provision of reporting and up-to-date insights. The Enterprise Architect, though often part of the Portfolio Leadership team, is highlighted in Figure 12 due to their specific responsibilities in establishing the portfolio's technology vision, strategy, and roadmap.

LPM processes encourage learning and feedback and facilitate strategic agility. In addition to value stream funding and the Kanban portfolio system, this also includes the regular measurement of portfolio performance against a set of objectives and key results (OKRs) and key performance indicators (KPIs). If the facts dictate that a pivot is required, the strategy is adjusted accordingly.

Implementing SAFe

Achieving the benefits of Lean-Agile development at scale requires significant and sustained effort. Successful transformation to full agility demands strong change leadership, persistent people practices, and a structured approach. The SAFe Implementation Roadmap (Figure 13) provides a step-by-step guide for adopting SAFe, leveraging proven change management strategies. To achieve the desired organizational change, leadership must "script the critical moves,"[8] as described by Chip and Dan Heath. When identifying those critical moves for adopting SAFe, many of the world's largest institutions have already gone down this path, and successful adoption patterns have emerged, as illustrated below.

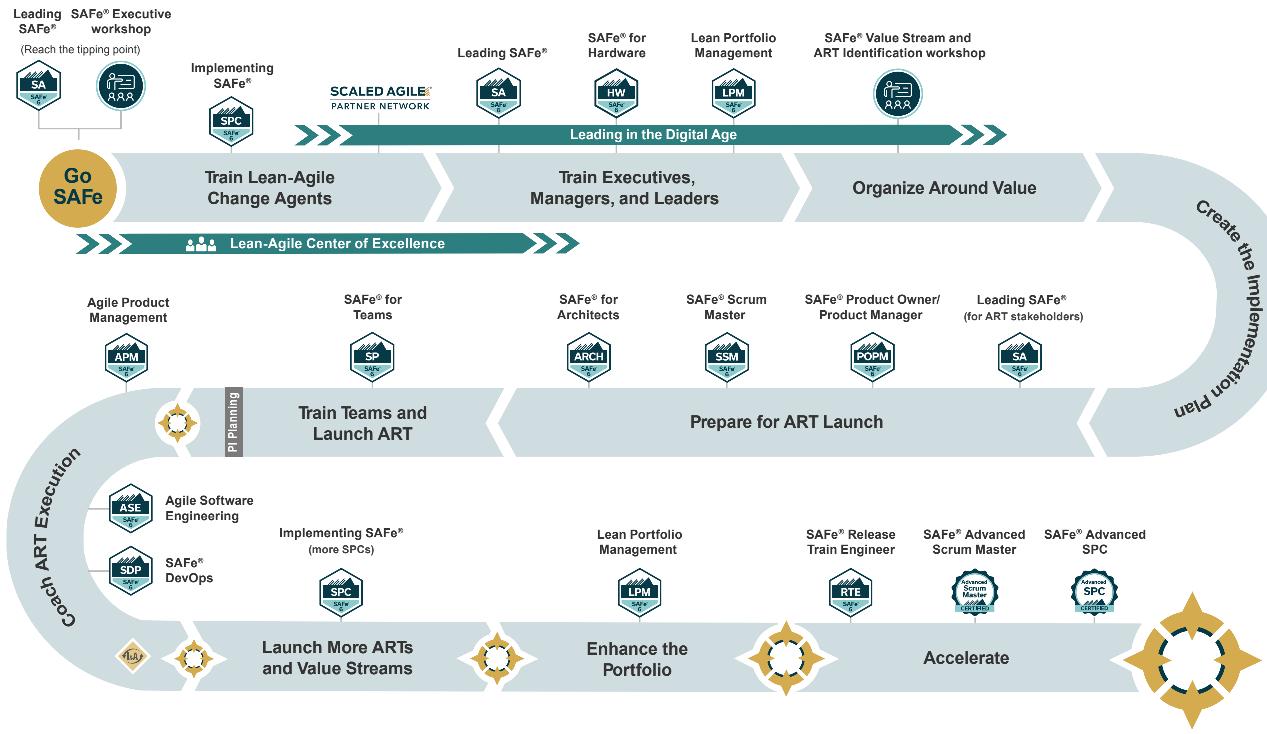

Figure 13. SAFe Implementation Roadmap

Conclusion

SAFe implementation demands continuous improvement and adaptation. While resistance to Lean-Agile principles and cultural shifts can be challenging, the rewards—faster time to market, better quality, higher productivity, and greater employee engagement—make it worthwhile. A long-term commitment to change, learning, and adjustment ensures successful adoption. By fostering collaboration, alignment, and adaptability, SAFe helps organizations meet market demands and achieve business goals efficiently.

[8] Chip Heath and Dan Heath, *Switch: How to Change Things When Change Is Hard* (New York: Crown Business, 2010)

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